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Key Points:

- We report a well-dated ~250 Ma paleomagnetic pole from the Mongolia Block
- The closure of both the Hegenshan and Solonker oceans at ~250 Ma marked the final disappearance of the Paleo-Asian Ocean

Supporting Information:

Supporting Information may be found in the online version of this article.

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Timing the Hegenshan Suture in the Central Asian Orogenic Belt: New Paleomagnetic and Geochronological Constraints From Southeastern Mongolia

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Abstract The timing of formation of the Hegenshan suture is crucial to understanding the evolution of the Central Asian Orogenic Belt (CAOB). Here we present a well-dated paleomagnetic pole (103 samples from 15 sites, 36.7°N/29.2°E, A_{95} = 3.3°) that passed a positive fold test, a reversal test, and a conglomerate test from the ~250 Ma upper Gunbayn Formation andesites in the southeastern Mongolia Block (MOB). The high-quality paleomagnetic database may demonstrate that the MOB drifted rapidly southward and collided with the Xilinhot–Songliao Block between 256 and 250 Ma, resulting in the closure of the Hegenshan Ocean at ~250 Ma. The Hegenshan suture and the Solonker suture were formed almost simultaneously and marked the final closure of the Paleo-Asian Ocean.

Plain Language Summary The Central Asian Orogenic Belt (CAOB) is the largest accretionary orogenic belt on earth since begin of the Phanerozoic. Since the Late Paleozoic, two prominent sutures, namely the Hegenshan and Solonker sutures, have played crucial roles in the tectonic evolution of the eastern CAOB, as they were associated with the closure of the Paleo-Asian Ocean (PAO). The timing of the formation of the Hegenshan suture, however, remains ambiguous due to the lack of reliable paleomagnetic data. In this study, we present accurately dated paleomagnetic constraints on the paleogeographic position of the Mongolia Block (MOB), located on the northern side of the Hegenshan suture, at ~250 Ma. By combining high-quality Late Permian–Early Triassic paleomagnetic data with geological evidence, we propose a new reconstruction that suggests the near-simultaneous closure of both the Hegenshan and Solonker oceans around 250 Ma. This significant event marks the final disappearance of the PAO and represents a fundamental stage in the establishment of the tectonic framework of the eastern CAOB at that time.

1. Introduction

The Central Asian Orogenic Belt (CAOB) is a huge accretionary and collisional orogenic belt, situated between the Baltica, Siberia, Tarim, and North China cratons (Liu et al., 2021; Xiao et al., 2015). The tectonic framework of the eastern CAOB was closely associated with the closure of the two principal branches of the Paleo-Asian Ocean (PAO) in the Late Paleozoic: the Hegenshan Ocean in the north and the Solonker Ocean in the south (Figure 1, Safonova et al., 2017; Zhang et al., 2018). Most studies on the Solonker suture suggest that it might have formed in the Late Permian and Early Triassic (Eizenhofer & Zhao, 2018; Ren et al., 2020; Zhang et al., 2014, 2023). It is commonly hypothesized that the suture represents the final closure of the PAO (Li, 2006; Liu et al., 2017; Wilde, 2015; Xiao et al., 2003). However, there are dissenting views among some researchers, as the timing of formation of the Hegenshan suture remains uncertain, with proposed ages ranging from Early Devonian to Middle Triassic (e.g., Domeier & Torsvik, 2014; Eizenhofer & Zhao, 2018; Liu et al., 2017, 2021; Song et al., 2015; Xiao et al., 2015, 2018; Xu et al., 2013; Zhang, Huang, Zhao, et al., 2021; Zhang et al., 2018). Therefore, determining when the Hegenshan suture was formed is crucial for comprehending the final closing time and location of the PAO, as well as the tectonic evolution of the CAOB.

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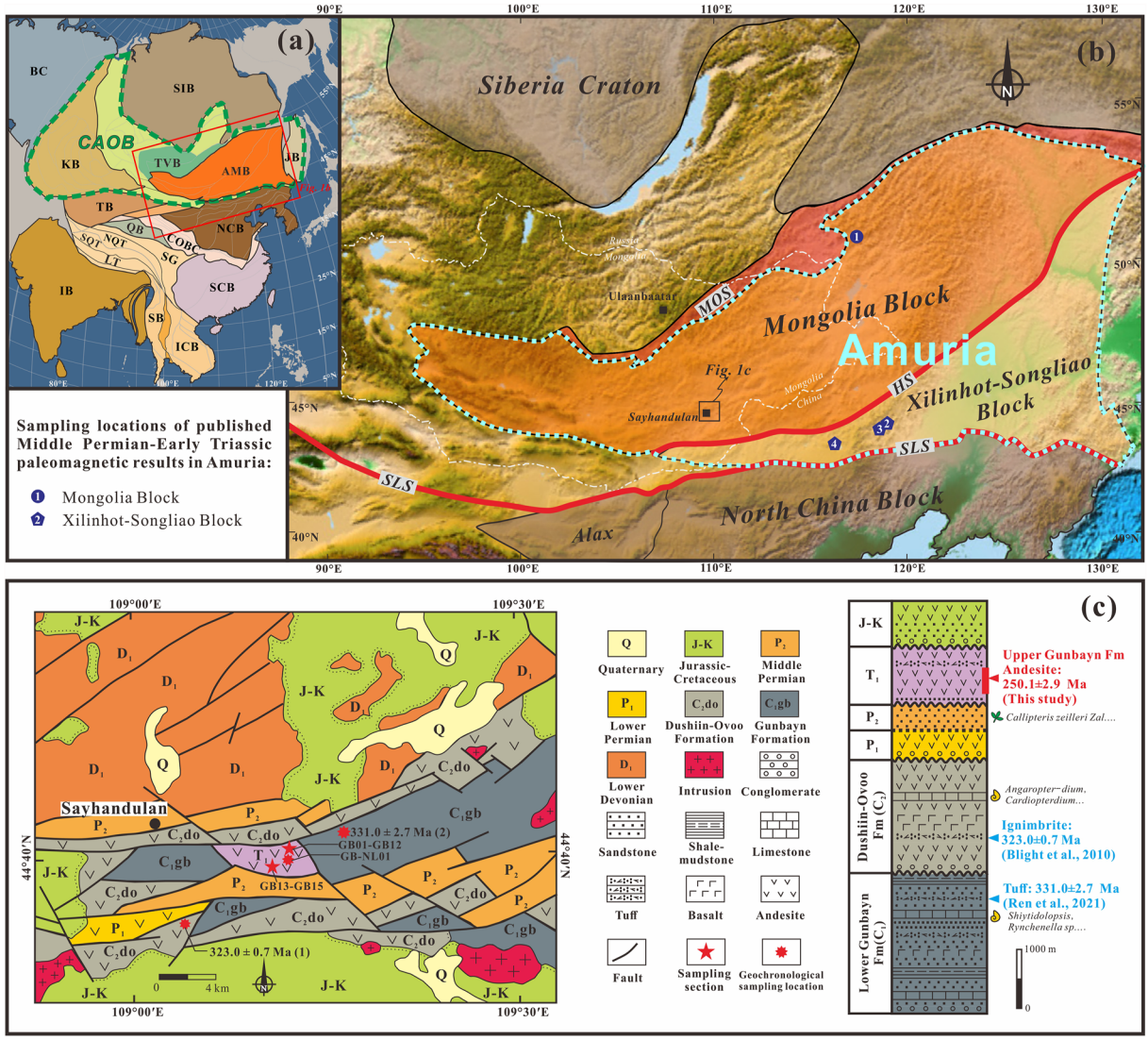


Figure 1. (a) Tectonic units of eastern Eurasia (modified from Ren et al. (2021)). Abbreviations: Amuria Block = AMB; CAOB = Central Asian Orogenic Belt; COBC = Central Orogenic Belt of China; BC = Baltica Craton; ICB = Indochina Block; IB = India Block; JB = Jiamusi–Bureya Block; KB = Kazakhstan Block; LT = Lhasa Terrane; MOS = Mongol–Okhotsk Suture; NCB = North China Block; NQT = North Qiangtang Terrane; QB = Qaidam Block; SCB = South China Block; SG = Songpan–Ganzi Belt; SB = Sibumasu Block; SIB = Siberia Craton; SQT = South Qiangtang Terrane; TB = Tarim Block. (b) Topographic and tectonic map of Eastern Central Asian Orogenic Belt, modified after (Liu et al., 2017). The numbers represent the middle Permian-Early Triassic paleomagnetic sampling locations: (1) Kravchinsky et al. (2002), (2)–(3) Didenko et al. (2016); (4) Ren et al. (2020). (c) Geological map and stratigraphic sequence in the Sayhandulan area of the southern Mongolia Block (modified from Tomurtogoo et al. (1998)).

To understand the process and timing of formation of the Hegenshan suture, direct comparison of the paleomagnetic poles for the Mongolia Block (MOB) and the Xilinhot–Songliao Block (XSB) on either side of the suture is required. Ren et al. (2022) and Zhang, Huang, Meert, et al. (2021) reported high-quality Early Permian paleomagnetic data from the XSB and MOB that suggested the existence of a wide Hegenshan Ocean between these two blocks during the Early Permian (Ren et al., 2022). Thus, comparison of paleomagnetic data from the Middle Permian to the Middle Triassic is necessary to study the closure time of this wide ocean. Although Middle Permian–Middle Triassic paleomagnetic data from the XSB on the southern side of this suture have been updated (Ren et al., 2020, 2022), coeval paleomagnetic data from the MOB are scarce. A single Late Permian pole from the northern MOB (Figure 1b) was reported by Kravchinsky et al. (2002), but that paleomagnetic result, which was obtained from clastic rocks, lacked analysis of inclination flattening (IF). Various studies have suggested that the paleomagnetic results for most clastic rocks from the eastern CAOB may have been affected by inclination

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shallowing (e.g., Ren et al., 2020; Zhang et al., 2018). Therefore, high-quality paleomagnetic data from the MOB are still required to constrain its paleogeographic location during the Middle Permian–Middle Triassic.

In this paper, we present new paleomagnetic and geochronological data for andesites from the upper Gunbayn Formation (Fm) of southeastern Mongolia, located in the MOB, with an age of ∼250 Ma. Our new data, combined with previously published Middle Permian–Middle Triassic paleomagnetic data and geological information, yield important insights into the paleogeographic relationship between the MOB and the XSB. Furthermore, we constrain the spatiotemporal evolution of the PAO, which played a key role in the formation of the CAOB.

2. Geological Setting and Sampling

The Amuria Block occupies the eastern CAOB between the Siberia Craton and the North China Block (NCB; Figure 1a), which was formed by the amalgamation of the MOB and XSB. Within the Amuria Block, the MOB occupies the northwestern region and is bounded by the Mongol–Okhotsk suture to the north and the Hegenshan suture to the south (Figure 1). This block comprises Precambrian terranes, massifs, island arcs, and accretionary wedges (e.g., the Ereendavaa terrane, Gobi–Altai terrane, Mandalovoo arc terrane, Gurvansayhan terrane, Erguna massif, and Xing'an massif), which assembled to form a single unit during the Early Paleozoic (Badarch et al., 2002; Ren et al., 2021; Xiao et al., 2018).

Paleomagnetic and geochronological samples were collected from volcanic rocks of the upper Gunbayn Fm in the central Mandalovoo zone in eastern Mongolia, MOB. This volcanogenic sequence is composed mainly of andesite, basalt, and volcaniclastic rock, with siltstone and sandstone interlayers (Tomurtogoo et al., 1998), and it is widely exposed in the Saiinshand area. This volcanic sequence has been assigned to the Early Carboniferous by previous geological mapping because it is similar to the rock assemblage of the surrounding lower Carboniferous strata (e.g., western Sayhandulan; Tomurtogoo et al., 1998). However, there is no fossil or radiometric isotope evidence to support this age determination, and the age of the upper Gunbayn Fm therefore remains unknown. The upper Gunbayn Fm is disconformably overlain by Early Cretaceous basalts (a plagioclase $^{40}Ar/^{39}Ar$ age of 131 ± 1 Ma; Graham et al., 2001) and disconformably underlain by the Early Carboniferous lower Gunbayn Fm (zircon U-Pb age of tuff at 331.0 ± 2.7 Ma; Ren et al., 2021). Therefore, we can infer only that the age of the volcanic sequence of the upper Gunbayn Fm formed between the Early Carboniferous and Early Cretaceous. To determine the eruption age of the volcanic rocks, we collected a fresh andesite sample (GB-NL01) for zircon U–Pb dating from our paleomagnetic sampling site GB06 (Figure 1c).

In this study, a total of 111 individual oriented andesite paleomagnetic samples from 15 sites (GB01–GB15) were collected from the upper Gunbayn Fm in the Sayhandulan area (Figure 1c and Figure S1 in Supporting Information S1). We obtained five to 10 cores at each paleomagnetic site. All of the paleomagnetic samples were obtained using a gasoline-powered portable drill with a water-cooled device and were oriented using both magnetic and sun compasses. The average of all differences was less than 2° between these two orientation methods, indicating that local magnetic disturbance is negligible. The attitudes of the andesitic lava flows were measured from interbedded volcaniclastic rocks (Figure S1 in Supporting Information S1).

3. Results

3.1. Zircon U-Pb Ages

Zircon grains from andesite sample GB-NL01 are mostly colorless and transparent. They show euhedral and prismatic shapes, display clear oscillatory zoning, and are typically 30–90 μm wide and 90–140 μm long in cathodoluminescence images (Figure S2a in Supporting Information S1; detailed methods are presented in Text S1 in Supporting Information S1). Th/U ratios are 0.57–1.30 (Table S1 in Supporting Information S1), implying a magmatic origin of the zircons. Ten analysis points yield a concordant age (Figure S2b in Supporting Information S1) and a weighted-mean $^{206}Pb/^{238}U$ age of 250.1 ± 2.9 Ma (MSWD = 0.068; Figure S2c in Supporting Information S1), which is interpreted as the crystallization age. Therefore, the eruption age of volcanic rocks of the upper Gunbayn (called new-Gunbayn) Fm in the corresponding paleomagnetic sampling section is ∼250 Ma.

3.2. Paleomagnetic Results

Paleomagnetic results of 103 paleomagnetic specimens from the 15 sampled sites have stable magnetic signals (Table S2 in Supporting Information S1 and Figures 2a–2d; detailed methods in Text S2.1 in Supporting

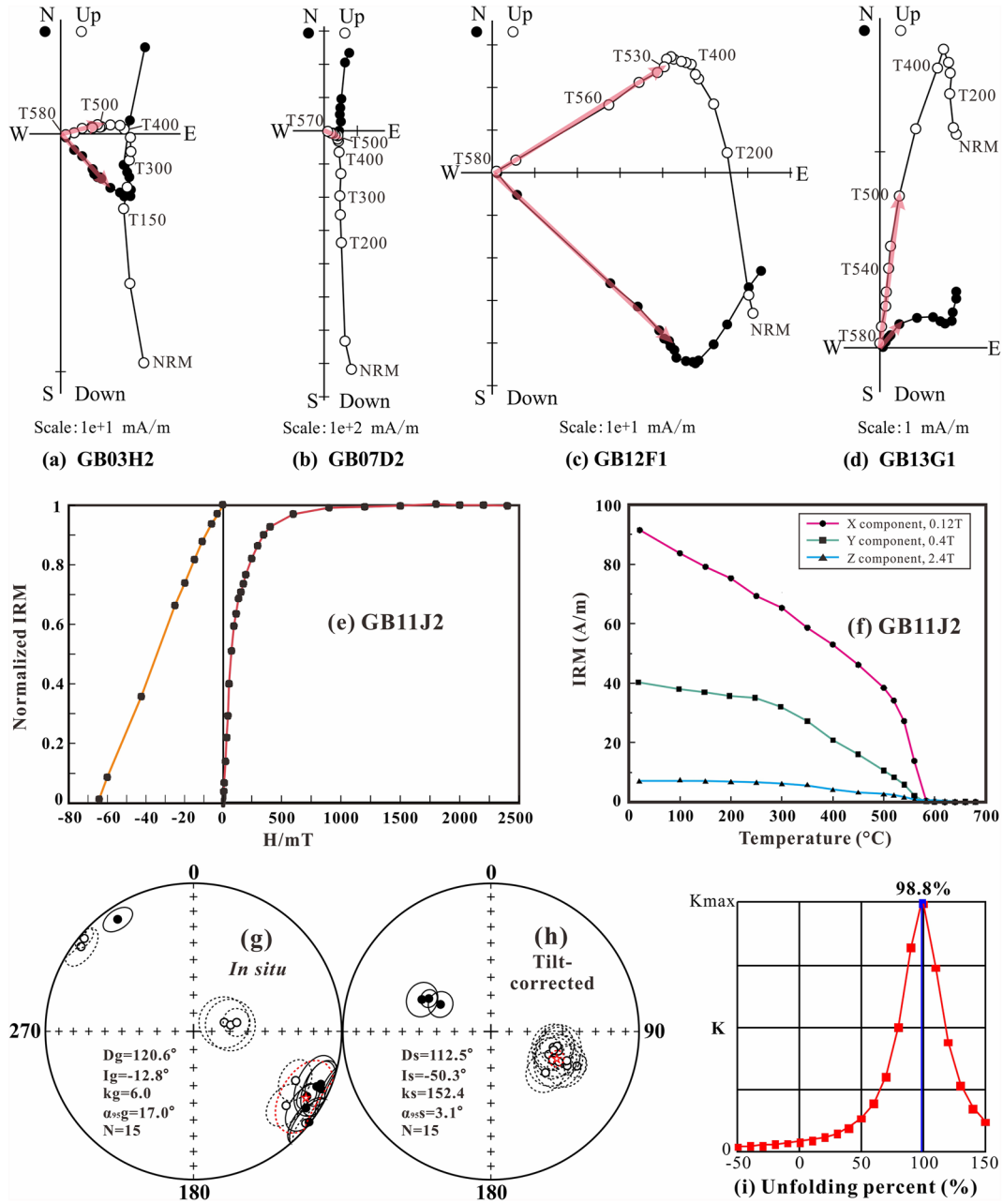


Figure 2. (a–d) Typical demagnetization characteristics of samples, in geographic coordinates. Solid/open symbols of the orthogonal plots represent the projections onto the horizontal/vertical plane. (e–f) Rock magnetic results of representative sample (GB11J2). (e) Isothermal remanent magnetization (IRM) and back-field demagnetization of saturation IRM acquisition curves. (f) Stepwise thermal demagnetization of the three orthogonal IRM components. (g–h) Equal-area stereographic projections of the site-mean directions for the new-Gunbayn Fm. Lower (upper) hemisphere directions are represented by solid (open) symbols; red stars indicate the formation-mean directions with 95% confidence circle of 15 sites. (i) Stepwise unfolding of the high-temperature components directions.

Information S1). The low-temperature components in situ, removed mostly below 300°C, are distributed around the local recent geomagnetic field (RGF; Figure S3 in Supporting Information S1) and represent a viscous remanent magnetization of the RGF. The stable high-temperature components (HTCs), typically isolated from 500 to 580°C (Figures 2a–2d), indicate that magnetite is the main magnetic carrier. These findings are supported by the results of rock-magnetic experiments (isothermal remanent magnetization (IRM), back-field demagnetization of saturation IRM, and Lowrie test; Text S2.2 in Supporting Information S1; Figures 2e and 2f). HTCs reveal dual-polarity

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directions (Figures 2g and 2h and Table S2 in Supporting Information S1), with magnetization directed northwest downward (southeast upward) as geomagnetically normal (reversed) polarity. The site-mean direction for the 15 sites is Dg = 120.6°, Ig = −12.8°, k = 6.0, and α_{95} = 17.0° in situ, and Ds = 112.5°, Is = −50.3°, k = 152.4, and α_{95} = 3.1° after tilt correction (Figures 2g and 2h and Table S2 in Supporting Information S1).

3.3. Reversal, Fold, and Conglomerate Tests

All site-mean HTCs pass a reversal test (McFadden & McElhinny, 1990) at the 95% confidence level (class B) with an angle between the two means of $\gamma_o = 4.2 < \gamma_{critical} = 7.8$. Both normal and reversal HTC groups share a common true mean from 1,000 bootstrapped Cartesian coordinates at the 95% confidence level (Tauxe, 2010), indicative of a positive reversal test.

The site-mean HTCs pass a fold test (McElhinny, 1964; McFadden, 1990) at the 95% and 99% confidence levels. Maximum clustering of the data occurs at 93%–106% untilting according to the bootstrap fold test (Tauxe & Watson, 1994). In the stepwise unfolding test, the value of precise parameter "k" reaches a maximum at 98.8% unfolding (Figure 2i; Watson & Enkin, 1993), which falls within the critical error interval of 90%–110% unfolding (Meert et al., 2020).

Seven paleomagnetic specimens from the new-Gunbayn Fm volcanic breccias (Figure S4a and S4b in Supporting Information S1) have stable magnetic signals (Figure S4c in Supporting Information S1). The low-temperature components in situ, removed below 300°C, are distributed around the local RGF (Figure S4d in Supporting Information S1) and represent a viscous remanent magnetization of the RGF. The stable HTCs decay toward the origin near the unblocking temperature of 550–580°C. The HTC directions from the different pebbles are randomly distributed (Figure S4e and S4f in Supporting Information S1). A random distribution test gives R = 3.38 (n = 7), which is significantly lower than the critical R-values of 4.18 and 4.89 at the 95% and 99% confidence levels, respectively (Watson, 1956), indicating a positive conglomerate test.

In summary, the positive reversal, fold, and conglomerate tests strongly indicate that the HTC represents primary magnetization.

3.4. Paleosecular Variation Test

Because a quickly cooled volcanic flow provides only a spot reading of geomagnetic field behavior (Meert et al., 2020), it is necessary to test whether small variations in the paleosecular variation (PSV) of the andesites of the new-Gunbayn Fm are averaged out. Calculations were made for 103 andesite samples from 15 sites. The PSV has been averaged out, as shown by the following lines of evidence: (a) A total of 15 paleomagnetic sites from at least 10 different lava flows are interbedded with many layers of sedimentary rocks; (b) the site-mean virtual geomagnetic poles (VGPs) show a Fisher distribution, as revealed by the positive result of the quantile–quantile plot test (Fisher et al., 1987; Mu = 0.820 < 1.207, Me = 0.502 < 1.094); and (c) the value of A_{95}, obtained from the VGPs of 103 lava specimens, is 2.7, which falls into the N-dependent A_{95} envelope with a 95% confidence interval (1.9, 4.4), as proposed by Deenen et al. (2011, 2014).

In summary, we obtained a mean paleomagnetic pole at 36.7°N/29.2°E (A_{95} = 3.3°) by averaging all of the site-mean VGPs for the ∼250 Ma new-Gunbayn Fm andesites (Table S2 in Supporting Information S1).

4. Discussion

4.1. Analysis of Middle Permian–Early Triassic Paleomagnetic Poles

Middle Permian–Early Triassic paleomagnetic data from the MOB and adjacent blocks (i.e., XSB, NCB and Siberia) were appraised by applying the newly revised seven-point data quality criterion system (Meert et al., 2020), the results of which are compiled in Table S3 in Supporting Information S1. Our data from the ∼250 Ma new-Gunbayn Fm volcanic rocks provide a new high-quality paleomagnetic pole for the MOB, as it meets all seven criteria (Table S3 in Supporting Information S1). For data from the XSB and NCB, we compiled reliable data by conducting reliability tests (i.e., field test and/or reversal test) recently reviewed by Ren et al. (2020). We excluded some data (e.g., Embleton et al., 1996; Qin et al., 2013; Wu et al., 1990) because no reliability test proved that they represent primary magnetization.

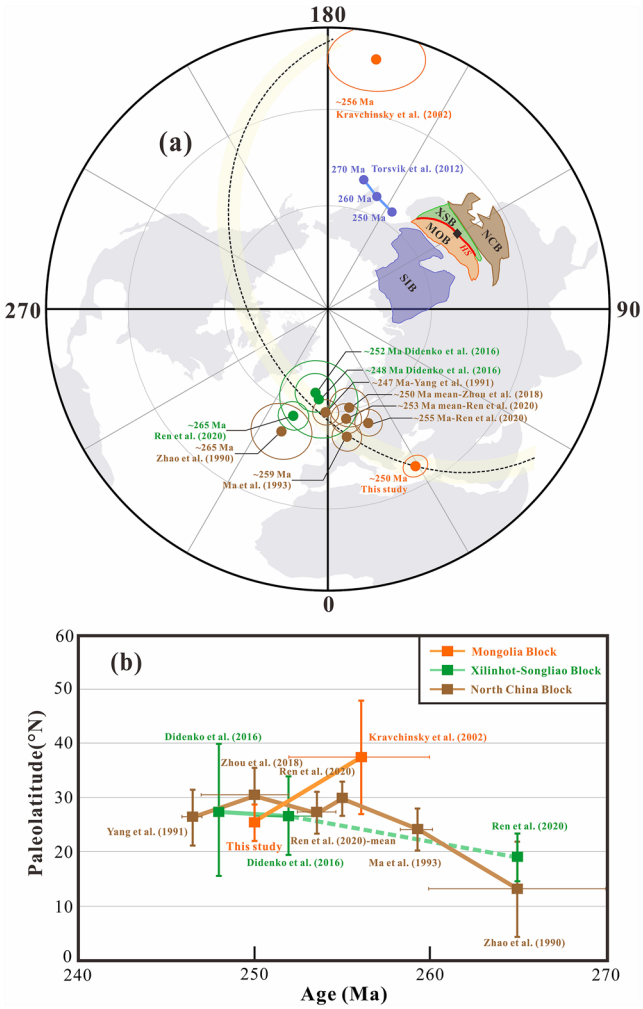


Figure 3. (a) Comparison of the middle Permian-Early Triassic paleomagnetic pole in the equal-area projection. The black diamond-shape is the reference site (45.7°N, 120.0°E) on the Hegenshan suture for calculating the small-circle with its 95% errors. Solid symbols are Lower Hemisphere projections. (b) The paleolatitude curves versus age. Data information see Table S3 in Supporting Information S1. The abbreviations: XSB = Xilinhot–Songliao Block; NCB = North China Block; MOB = Mongolia Block; SIB = Siberia; HS = Hegenshan Suture.

Clastic rocks commonly carry a detrital remanent magnetization that may be affected by IF during deposition and compaction (King, 1955). Most studies reporting paleomagnetic data for clastic rocks have used a blanket flattening factor (f = 0.6) to correct their inclinations (e.g., Torsvik et al., 2012; Van der Voo et al., 2015) via the elongation/inclination (E/I) correction proposed by Tauxe et al. (2008). In our compiled paleomagnetic data set for the MOB, XSB, and NCB (Table S3 in Supporting Information S1), some previous data from classic rocks have not been tested or corrected for inclination shallowing (i.e., Yang et al., 1991; Kravchinsky et al., 2002; Ren et al., 2020-mean; Ma et al., 1993). In this study, published data from fine-grained clastic rocks were corrected using a flattening factor of f = 0.6, for the following reasons: (a) Paleomagnetic HTC directions obtained from upper Permian Belektuy Fm fine-grained sandstones show a significant east–west elongated distribution (see Figure 4a in Kravchinsky et al., 2002) and do not follow a Fisher distribution, as revealed by the negative result of a quantile–quantile plot test (Fisher et al., 1987; Mu = 1.244 > 1.207, Me = 0.462 < 1.094). However, the data follow a Fisher distribution after IF correction (f = 0.6), as revealed by the positive result of the quantile–quantile plot test (Fisher et al., 1987; Mu = 0.965 < 1.207, Me = 0.560 < 1.094). (b) Late Permian data obtained from fine-grained sandstones–clays of the upper Shihezi and Shiqianfeng fms in the Ordos Basin (reviewed in Ren et al., 2020) yield a paleolatitude of ~18.3°N with a reference site (45.7°N, 120.0°E) lower than that of the ~30.7°N obtained from the upper Qingfengshan Fm volcanic rocks (Ren et al., 2020). (c) this blanket 0.6 correction is actually based on abundant real data in central Asia (e.g., in the North Tianshan Zone, Kirscher et al., 2014; Tuva-Mongol Zone, Ren et al., 2021). The factor of 0.6 is also equal to the median of the mean range of inclination shallowing for magnetite and hematite, as summarized by Kodama (2012). In After IF correction with f = 0.6 for the clastic rocks, the paleolatitudes from data for both clastic and volcanic rocks were similar. These results suggest that paleomagnetic data from fine-grained clastic rocks of the MOB, XSB, and NCB have likely been affected by IF and that the blanket f = 0.6 is reasonable for correcting the inclination. However, we did not use f = 0.6 to correct the Middle Triassic data from the Zhifang Fm sandstones (Yang et al., 1991), as the site-mean VGPs follow a Fisher distribution, as shown by the positive result of a quantile–quantile plot test (Fisher et al., 1987; Mu = 0.898 < 1.207, Me = 0.893 < 1.094), but they do not follow a Fisher distribution after applying an IF correction. Moreover, the paleolatitude obtained from the data of Yang et al. (1991) was consistent with the coeval paleolatitude from clastic rocks (Zhou et al., 2018) after IF correction (f = 0.6).

To better analyze and compare the Late Permian–Early Triassic paleomagnetic poles for the MOB, XSB, and NCB, we observed all of the poles using a mixed mean between fixed poles along a common small circle (Figure 3a) and translated these poles into a graph of paleolatitude evolution (Figure 3b) at a reference site (45.7°N, 120.0°E) along the Hegenshan suture. Currently, there is only one reliable paleomagnetic paleopole from the Late Permian for the MOB (Kravchinsky et al., 2002), and this pole differs significantly from the coeval poles for the XSB and NCB (Figure 3a). We compared the expected site-mean directions (with respect to the reference site) between clastic rocks from Kravchinsky et al. (2002; with an IF correction, f = 0.6) and volcanic rocks from Ren et al. (2020) by using a significance test from Tauxe (2010). Our analysis, based on 1,000 bootstrapped Cartesian coordinates, showed no significant match of the two data sets at the 95% confidence level (Figure S5 in Supporting Information S1), indicating a statistically significant difference between them. A considerable latitudinal difference of 6°–10° for the reference site is observed between the MOB and XSB (Figure 3b). By the Early Triassic (~250 Ma), the poles for the MOB, XSB, and NCB were located along the small circle (Figure 3a), suggesting that there was no significant latitudinal difference among them (Figure 3b).

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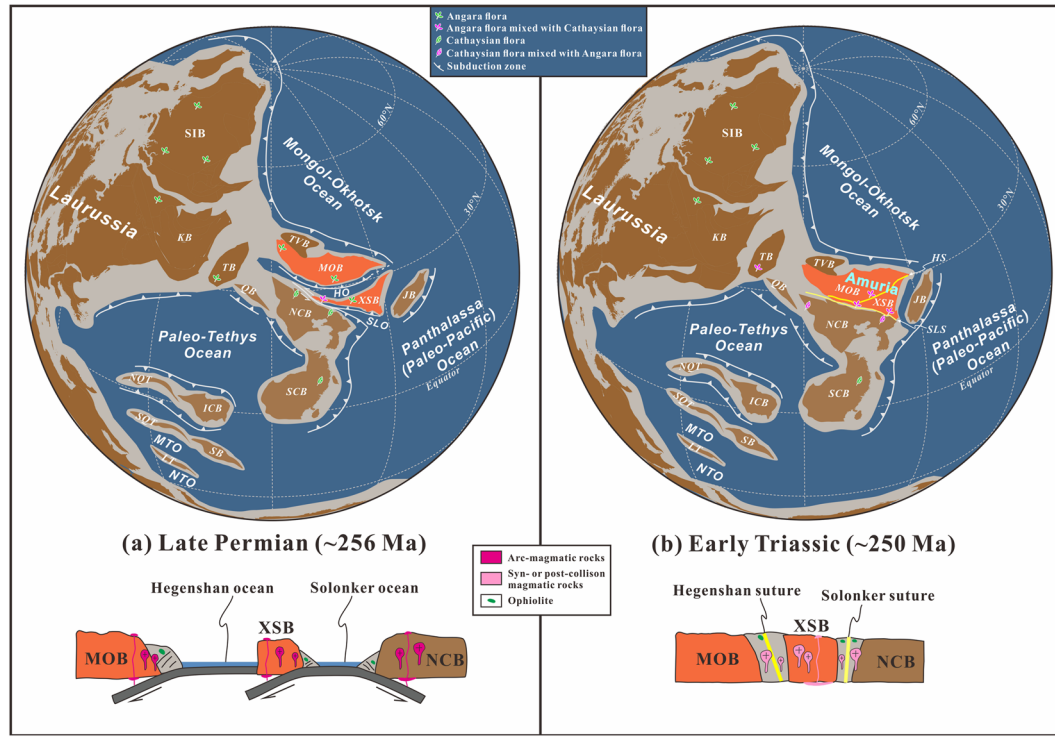


Figure 4. Schematic paleogeographic reconstruction of the Northeast Asian blocks and Gondwana, Laurussia and Siberia from ~256 to ~250 Ma using *Gplates* software (Boyden et al., 2011). Gondwana, Laurussia, and Siberia are reconstructed following the relative Euler rotation parameters given by Domeier and Torsvik (2014). The Euler rotation parameters of the Tethyan tectonic domain (e.g., the South China, North and South Qiantang, Lhasa, Indochina, and Sibumasu blocks) follow Tang et al. (2022). The model of the subduction-collision of the Paleo-Asian Ocean (Hegenshan and Solonker oceans) corresponds to the reconstructions. For abbreviations see caption of Figure 1, and HO = Hegenshan Ocean; HS = Hegenshan Suture; MOB = Mongolia Block; MTO = Meso–Tethys Ocean; NTO = Neo-Tethys Ocean; SLO = Solonker Ocean; SLS = Solonker Suture; XSB = Xilinhot-Songliao Block.

4.2. Implications for the Paleogeographic Evolution of the Eastern CAOB

To explain the formation process of the eastern CAOB, we propose a new paleogeographic model of the MOB and XSB and a new subduction–collision process for the evolution of the PAO (including the Hegenshan and Solonker oceans) by integrating and interpreting current paleomagnetic and geological information (Figure 4).

Our new paleomagnetic data show that during the Late Permian (~256 Ma), there was a 6°–10° difference in paleolatitude between the MOB and the XSB (Figure 3b), which suggests that there has been 660–1,100 km of lithospheric shortening between them since ~256 Ma. This inference is supported by the following lines of geological evidence: (a) Late Permian marine to shallow-marine strata (e.g., the Suolun Fm) are extensively distributed in the Xilinhot, western Ujimqin, Hegenshan, and Suolun areas on both sides of the Hegenshan suture, as reliably demonstrated by the fossil record (Eizenhofer & Zhao, 2018; Tian et al., 2016); (b) subduction-related magmatism is recorded in the southern margin of the MOB (e.g., the 315–250 Ma Dashizhai Fm volcanic rocks; Song et al., 2015; Eizenhofer & Zhao, 2018; Figure S2a in Supporting Information S1); (c) the Hegenshan ophiolite mélanges formed during the Early Permian (ages of ~295 and ~298 Ma for gabbros and mafic dikes, respectively) in an oceanic rift basin and were finally emplaced during the Early Triassic (Xiao et al., 2018); and (d) the peak ages of detrital zircons from Middle–Late Permian Zhesi Fm sandstones differ between the northern and southern sides of the Hegenshan suture (Eizenhofer & Zhao, 2018). Therefore, a limited Hegenshan Ocean basin, as the northern branch of the PAO, should have existed between the MOB and the XSB at ~256 Ma (Figure 4a). Due to the extensive development of Permian subduction-related magmatic rocks in the eastern MOB, the Hegenshan Ocean should extend northward (Figure 4a), not just between the MOB and Xilinhot Block (Zhang, Huang, Meert, et al., 2021). Combining our results and interpretations with the paleomagnetic models of Zhang, Huang, Zhao, et al. (2021) and

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Ren et al. (2022), we speculate that the Hegenshan Ocean may have closed for the first time prior to ~300 Ma and then reopened during the Early Permian. However, due to the scarcity of Early–Middle Permian paleomagnetic data from the MOB, it is not clear whether the Hegenshan Ocean underwent multi-stage tectonic evolution, and this needs to be investigated in future studies. The other southern branch of the PAO, the small wedge-shaped Solonker Ocean, was formed by the western NCB colliding with the XSB (Figure 4a), as inferred from paleomagnetic and geological data (Ren et al., 2020). The Angara Flora therefore began to mix with the Cathaysia Flora (Figure 4a); for example, some typical Cathaysia Flora (or Gondwana-Cathaysia Flora) has been discovered in late Middle Permian strata and Late Permian strata in southeastern Mongolia in the western XSB (Naugolnykh & Uranbileg, 2018; Sun, 1999). However, possibly because of the geographic barrier of the Hegenshan Ocean, the migration and mixing of the Angara and Cathaysia floras did not occur in the MOB (Figure 4a).

From the Late Permian (~256 Ma) to the Early Triassic (~250 Ma), the XSB did not undergo any substantial plate movement, but the MOB moved rapidly southward (Figure 3b), which completed the collision between the MOB and the XSB and finally closed the Hegenshan Ocean (Figure 4b). This interpretation is supported by the following geological observations: (a) Early Triassic terrestrial strata are distributed on both the southern and northern sides of the Hegenshan suture; (b) similar detrital zircon age spectra are found in Early Triassic sandstones (i.e., the Linxi Fm) of both the MOB and XSB (Han et al., 2012); (c) syn-collisional (S-type) granitic magmatism occurred in the XSB, as represented, for example, by the Early Triassic Halatu granite suite (Chen et al., 2009); and (d) Early Triassic compressive deformation (e.g., folding and thrusting structures) is recorded in Early Paleozoic strata of the northern Solonker suture (Li, 2006; Xiao et al., 2018). The amalgamation of the MOB and XSB formed the unified Amuria Block during the Early Triassic (~250 Ma; Figure 4b). The Amuria Block did not include the Jiamusi Block (JB) at this time. The similar detrital zircon age spectra and the spatial–temporal distribution of granitoids suggest that the JB was connected to the XSB during the early Paleozoic (Liu et al., 2021; Xu et al., 2019) but then broke away during the Late Permian–Early Triassic and subsequently reconnected during the Late Mesozoic (Liu et al., 2017, 2019, 2021; Ren et al., 2016, 2018). From the Late Permian to the Early Triassic, the Solonker Ocean between the XSB and NCB underwent a scissor-like closure process, and the XSB connected with the NCB during the Early Triassic (Figure 4b). Unconstrained mixing of the Angaran and Cathaysian floras took place across the entire Amuria Block and the NCB (Figure 4b). Moreover, both the Hegenshan and Solonker oceans underwent final closure at ~250 Ma, marking the final disappearance of the PAO. Therefore, we conclude that the Hegenshan Ocean and the Solonker Ocean closed almost simultaneously at ~250 Ma, marking the final disappearance of the PAO at this time. This is contrary to the previous view that the closure of the Heganshan Ocean was significantly earlier than the Solonker Ocean. Our reconstruction shows that the unified Amuria was amalgamated with the NCB at ~250 Ma, implying that the fundamental tectonic framework of the eastern CAOB was established at that time (Figure 4b).

5. Conclusions

We obtained new accurately dated (~250 Ma) paleomagnetic data from the new-Gunbayn Fm andesites that meet all criteria of the modified seven-point quality criterion system (Meert et al., 2020). The corresponding ~250 Ma paleomagnetic pole for the MOB was located at 36.7°N/29.2°E (A_{95} = 3.3°). Comparison of high-quality Late Permian–Early Triassic paleomagnetic poles for the MOB and XSB shows substantial latitudinal convergence after ~256 Ma, revealing the existence of a limited Hegenshan Ocean between these two blocks. From 256 to 250 Ma, the MOB drifted rapidly southward to collide with the XSB, resulting in the closure of the Hegenshan Ocean basin and the final formation of the unified Amuria Block at ~250 Ma. At the same time, the scissor-like collision of the XSB with the NCB from west to east resulted in the closure of the Solonker Ocean. The near-simultaneous closure of both the Hegenshan and Solonker oceans at ~250 Ma marked the final disappearance of the PAO and the fundamental establishment of the tectonic framework of the eastern CAOB.

Data Availability Statement

Data to support this study are available in Zenodo (https://doi.org/10.5281/zenodo.7473261).

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References

Badarch, G., Cunningham, W. D., & Windley, B. F. (2002). A new terrane subdivision for Mongolia: Implications for the phanerozoic crustal growth of Central Asia. *Journal of Asian Earth Sciences*, *21*(1), 87–110. https://doi.org/10.1016/s1367-9120(02)00017-2

Boyden, J. A., Müller, R. D., Gurnis, M., Torsvik, T. H., Clark, J. A., Turner, M., et al. (2011). *Next-generation plate-tectonic reconstructions using GPlates*. Cambridge University Press.

Chen, B., Jahn, B., & Tian, W. (2009). Evolution of the Solonker suture zone: Constraints from zircon U–Pb ages, Hf isotopic ratios and whole-rock Nd–Sr isotope compositions of subduction-and collision-related magmas and forearc sediments. *Journal of Asian Earth Sciences*, *34*(3), 245–257. https://doi.org/10.1016/j.jseaes.2008.05.007

Deenen, M. H., Langereis, C. G., van Hinsbergen, D. J., & Biggin, A. J. (2011). Geomagnetic secular variation and the statistics of palaeomagnetic directions. *Geophysical Journal International*, *186*(2), 509–520. https://doi.org/10.1111/j.1365-246x.2011.05050.x

Deenen, M. H., Langereis, C. G., van Hinsbergen, D. J., & Biggin, A. J. (2014). Erratum: Geomagnetic secular variation and the statistics of palaeomagnetic directions. *Geophysical Journal International*, *197*(1), 643. https://doi.org/10.1093/gji/ggu021

Domeier, M., & Torsvik, T. H. (2014). Plate tectonics in the late paleozoic. *Geoscience Frontiers*, *5*(3), 303–350. https://doi.org/10.1016/j.gsf.2014.01.002

Eizenhofer, P., & Zhao, G. (2018). Solonker Suture in East Asia and its bearing on the final closure of the eastern segment of the Palaeo-Asian Ocean.

Embleton, B. J. J., Mcelhinny, M. W., Ma, X., Zhang, Z., & Zheng, X. L. (1996). Permo-Triassic magnetostratigraphy in China: The type section near Taiyuan, Shanxi Province, North China. *Geophysical Journal of the Royal Astronomical Society*, *126*(2), 382–388. https://doi.org/10.1111/j.1365-246x.1996.tb05298.x

Fisher, N., Lewis, T., & Embleton, B. (1987). *Statistical analysis of spherical data*. Cambridge University Press.

Graham, S. A., Hendrix, M. S., Johnson, C. L., Badamgarav, D., Badarch, G., Amory, J., et al. (2001). Sedimentary record and tectonic implications of Mesozoic rifting in Southeast Mongolia. *Geological Society of America Bulletin*, *113*(12), 1560–1579. https://doi.org/10.1130/0016-7606(2001)113<1560:sratio>2.0.co;2

Han, G., Liu, Y., Neubauer, F., Jin, W., Genser, J., Ren, S., et al. (2012). LA-ICP-MS U–Pb dating and Hf isotopic compositions of detrital zircons from the Permian sandstones in Da Xing'an Mountains, NE China: New evidence for the eastern extension of the Erenhot–Hegenshan suture zone. *Journal of Asian Earth Sciences*, *49*, 249–271. https://doi.org/10.1016/j.jseaes.2011.11.011

King, R. F. (1955). The remanent magnetism of artificially deposited sediments. *Monthly Notices of the Royal Astronomical Society*, *7*(7), 115–134. https://doi.org/10.1111/j.1365-246x.1955.tb06558.x

Kirscher, U., Bilardello, D., Mikolaichuk, A., & Bachtadse, V. (2014). Correcting for inclination shallowing of early Carboniferous sedimentary rocks from Kyrgyzstan—Indication of stable subtropical position of the North Tianshan Zone in the mid-late Palaeozoic. *Geophysical Journal International*, *198*(2), 1000–1015. https://doi.org/10.1093/gji/ggu177

Kodama, K. P. (2012). *Paleomagnetism of sedimentary rocks: Process and interpretation*. John Wiley & Sons.

Kravchinsky, V. A., Cogné, J.-P., Harbert, W. P., & Kuzmin, M. I. (2002). Evolution of the Mongol-Okhotsk Ocean as constrained by new palaeomagnetic data from the Mongol-Okhotsk Suture Zone, Siberia. *Geophysical Journal International*, *148*(1), 34–57. https://doi.org/10.1046/j.1365-246x.2002.01557.x

Li, J. Y. (2006). Permian geodynamic setting of Northeast China and adjacent regions: Closure of the Paleo-Asian Ocean and subduction of the Paleo-Pacific Plate. *Journal of Asian Earth Sciences*, *26*(3), 207–224. https://doi.org/10.1016/j.jseaes.2005.09.001

Liu, Y., Feng, Z., Jiang, L., Jin, W., Li, W., Guan, Q., et al. (2019). Ophiolite in the eastern Central Asian Orogenic Belt, NE China. *Acta Petrologica Sinica*, *35*(10), 3017–3047. https://doi.org/10.18654/1000-0569/2019.10.05

Liu, Y., Li, W., Feng, Z., Wen, Q., Neubauer, F., & Liang, C. (2017). A review of the Paleozoic tectonics in the eastern part of Central Asian Orogenic Belt. *Gondwana Research*, *43*, 123–148. https://doi.org/10.1016/j.gr.2016.03.013

Liu, Y., Li, W., Ma, Y., Feng, Z., Guan, Q., Li, S., et al. (2021). An Orocline in the eastern Central Asian Orogenic Belt. *Earth-Science Reviews*, *221*, 103808. https://doi.org/10.1016/j.earscirev.2021.103808

Ma, X., Xing, L., Yang, Z., Xu, S., & Zhang, J. (1993). Paleomagnetic study since Late Paleozoic in the Ordos basin. *Acta Geophysica Sinica*, *36*(1), 68–79.

McElhinny, M. (1964). Statistical significance of the fold test in palaeomagnetism. *Geophysical Journal International*, *8*(3), 338–340. https://doi.org/10.1111/j.1365-246x.1964.tb06300.x

McFadden, P. (1990). A new fold test for palaeomagnetic studies. *Geophysical Journal International*, *103*(1), 163–169. https://doi.org/10.1111/j.1365-246x.1990.tb01761.x

McFadden, P., & McElhinny, M. (1990). Classification of the reversal test in palaeomagnetism. *Geophysical Journal International*, *103*(3), 725–729. https://doi.org/10.1111/j.1365-246x.1990.tb05683.x

Meert, J. G., Pivarunas, A. F., Evans, D. A., Pisarevsky, S. A., Pesonen, L. J., Li, Z.-X., et al. (2020). The magnificent seven: A proposal for modest revision of the quality index. *Tectonophysics*, *790*, 228549. https://doi.org/10.1016/j.tecto.2020.228549

Naugolnykh, S. V., & Uranbileg, L. (2018). A new discovery of Glossopteris in southeastern Mongolia as an argument for distant migration of Gondwanan plants. *Journal of Asian Earth Sciences*, *154*, 142–148. https://doi.org/10.1016/j.jseaes.2017.11.039

Qin, H., Li, Y., Huang, S., Cai, S., & Ren, S. (2013). Palaeomagnetic investigation of Permain sandstone in Taohaiyingzi area of Inner Mongolia and its tectonic significance. *Geological Bulletin of China*, *32*(2–3), 388–398.

Ren, Q., Zhang, S., Gao, Y., Zhao, H., Wu, H., Yang, T., & Li, H. (2020). New middle–Late Permian paleomagnetic and geochronological results from Inner Mongolia and their paleogeographic implications. *Journal of Geophysical Research: Solid Earth*, *125*(7), e2019JB019114. https://doi.org/10.1029/2019jb019114

Ren, Q., Zhang, S., Hou, M., Wu, H., Yang, T., Li, H., & Chen, A. (2022). New early permian paleomagnetic and geochronological data from the Xilinhot–Songliao Block and their implications for the relationship between the Paleo-Asian Ocean and the paleo-Tethys ocean. *Geophysical Research Letters*, *49*(19), e2022GL100462. https://doi.org/10.1029/2022gl100462

Ren, Q., Zhang, S., Sukhbaatar, T., Zhao, H., Wu, H., Yang, T., et al. (2021). Did the Boreal Realm extend into the equatorial region? New paleomagnetic evidence from the Tuva–Mongol and Amuria blocks. *Earth and Planetary Science Letters*, *576*, 117246. https://doi.org/10.1016/j.epsl.2021.117246

Ren, Q., Zhang, S., Wu, H., Liang, Z., Miao, X., Zhao, H., et al. (2016). Further paleomagnetic results from the ~155 Ma Tiaojishan formation, Yanshan Belt, North China, and their implications for the tectonic evolution of the Mongol–Okhotsk suture. *Gondwana Research*, *35*, 180–191. https://doi.org/10.1016/j.gr.2015.05.002

19448007, 2023, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2023GL104881 by Curtin University Library, Wiley Online Library on [07/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Ren, Q., Zhang, S., Wu, Y., Yang, T., Gao, Y., Turbold, S., et al. (2018). New Late Jurassic to Early Cretaceous paleomagnetic results from North China and southern Mongolia and their implications for the evolution of the Mongol-Okhotsk suture. *Journal of Geophysical Research: Solid Earth*, *123*(12), 10,370–10,398. https://doi.org/10.1029/2018jb016703

Safonova, I., Kotlyarov, A., Krivonogov, S., & Xiao, W. (2017). Intra-oceanic arcs of the Paleo-Asian Ocean. *Gondwana Research*, *50*(3), 167–194. https://doi.org/10.5800/gt-2017-8-3-0287

Song, S., Wang, M. M., Xu, X., Wang, C., Niu, Y., Allen, M. B., & Su, L. (2015). Ophiolites in the Xing'an-Inner Mongolia accretionary belt of the CAOB: Implications for two cycles of seafloor spreading and accretionary orogenic events. *Tectonics*, *34*(10), 2221–2248. https://doi.org/10.1002/2015tc003948

Sun, K. (1999). Origin, evolution and extinction of Cathaysia Flora. *Chinese Science Bulletin*, *44*(2), 100–108. https://doi.org/10.1007/bf02884727

Tang, M., Ren, Q., Hou, M., Li, Z., Du, W., Chen, A., et al. (2022). Digital paleogeographic reconstruction of the eastern Tethyan tectonic domain from the Middle Permian to the Middle Triassic. *Geosystems and Geoenvironment*, 100127. https://doi.org/10.1016/j.geogeo.2022.100127

Tauxe, L. (2010). *Essentials of paleomagnetism*. University of California Press.

Tauxe, L., Kodama, K. P., & Kent, D. V. (2008). Testing corrections for paleomagnetic inclination error in sedimentary rocks: A comparative approach. *Physics of the Earth and Planetary Interiors*, *169*(1–4), 152–165. https://doi.org/10.1016/j.pepi.2008.05.006

Tauxe, L., & Watson, G. (1994). The fold test: An Eigen analysis approach. *Earth and Planetary Science Letters*, *122*(3–4), 331–341. https://doi.org/10.1016/0012-821x(94)90006-x

Tian, S., Li, Z., Zhang, Y., Gong, Y., Zhai, D., & Wang, M. (2016). Late Carboniferous-Permian tectono-geographical conditions and development in eastern Inner Mongolia and adjacent areas. *Acta Geologica Sinica*, *90*(4), 688–707.

Tomurtogoo, O., Byamba, J., Badarch, G., Minjin, C., Orolmaa, D., Khosbayar, P., & Chuluun, D. (1998). *Geologic map of Mongolia, Scale 1: 1,000,000 and summary*. Mineral Resources Authority of Mongolia.

Torsvik, T. H., Van der Voo, R., Preeden, U., Mac Niocaill, C., Steinberger, B., Doubrovine, P. V., et al. (2012). Phanerozoic polar wander, palaeogeography and dynamics. *Earth-Science Reviews*, *114*(3–4), 325–368. https://doi.org/10.1016/j.earscirev.2012.06.007

Van der Voo, R., van Hinsbergen, D. J., Domeier, M., Spakman, W., & Torsvik, T. H. (2015). Latest Jurassic–earliest Cretaceous closure of the Mongol-Okhotsk Ocean: A paleomagnetic and seismological-tomographic analysis. *Geological Society of America Special Paper*, *513*, 589–606.

Watson, G. (1956). A test for randomness of directions. *Geophysical Supplements to the Monthly Notices of the Royal Astronomical Society*, *7*(4), 160–161. https://doi.org/10.1111/j.1365-246x.1956.tb05561.x

Watson, G. S., & Enkin, R. J. (1993). The fold test in paleomagnetism as a parameter estimation problem. *Geophysical Research Letters*, *20*(19), 2135–2137. https://doi.org/10.1029/93gl01901

Wilde, S. A. (2015). Final amalgamation of the Central Asian Orogenic Belt in NE China: Paleo-Asian Ocean closure versus Paleo-Pacific plate subduction—A review of the evidence. *Tectonophysics*, *662*, 345–362. https://doi.org/10.1016/j.tecto.2015.05.006

Wu, H., Zhu, R., Liu, C., & Chang, C. (1990). Paleomagnetic observations in North China Block: From Late Paleozoic to Triassic. *Acta Geophysica Sinica*, *33*, 694–701.

Xiao, W., Brian, F. W., & Hao, J. (2003). Accretion leading to collision and the Permian Solonker Suture, Inner Mongolia, China: Termination of the central Asian Orogenic Belt. *Tectonics*, *22*(6), 1069. https://doi.org/10.1029/2002TC001484

Xiao, W., Windley, B. F., Han, C., Liu, W., Wan, B., Zhang, J. e., et al. (2018). Late Paleozoic to early Triassic multiple roll-back and oroclinal bending of the Mongolia collage in Central Asia. *Earth-Science Reviews*, *186*, 94–128. https://doi.org/10.1016/j.earscirev.2017.09.020

Xiao, W., Windley, B. F., Sun, S., Li, J., Huang, B., Han, C., et al. (2015). A tale of amalgamation of three Permo-Triassic collage systems in Central Asia: Oroclines, sutures, and terminal accretion. *Annual Review of Earth and Planetary Sciences*, *43*(1), 477–507. https://doi.org/10.1146/annurev-earth-060614-105254

Xu, B., Charvet, J., Chen, Y. C., Zhao, P., & Shi, G. (2013). Middle Paleozoic convergent orogenic belts in western Inner Mongolia (China): Framework, kinematics, geochronology and implications for tectonic evolution of the Central Asian orogenic Belt. *Gondwana Research*, *23*(4), 1342–1364. https://doi.org/10.1016/j.gr.2012.05.015

Xu, W., Sun, C., Tang, J., Luan, J., & Wang, F. (2019). Basement nature and tectonic evolution of the Xing'an-Mongolian Orogenic Belt. *Earth Science*, *44*(5), 1620–1646.

Yang, Z., Ma, X., Besse, J., Courtillot, V., Xing, L., Xu, S., & Zhang, J. (1991). Paleomagnetic results from Triassic sections in the Ordos basin, North China. *Earth and Planetary Science Letters*, *104*(2–4), 258–277. https://doi.org/10.1016/0012-821x(91)90208-y

Zhang, D., Huang, B., Meert, J. G., Zhao, G., Zhao, J., & Zhao, Q. (2021). Micro-Blocks in NE Asia amalgamated into the Unified Amuria Block by ∼300 Ma: First paleomagnetic evidence from the Songliao Block, NE China. *Journal of Geophysical Research: Solid Earth*, *126*(10), e2021JB022881. https://doi.org/10.1029/2021jb022881

Zhang, D., Huang, B., Zhao, G., Meert, J. G., Williams, S., Zhao, J., & Zhou, T. (2021b). Quantifying the extent of the Paleo-Asian Ocean during the late carboniferous to early Permian. *Geophysical Research Letters*, *48*(15), e2021GL094498. https://doi.org/10.1029/2021gl094498

Zhang, D., Huang, B., Zhao, J., Meert, J. G., Zhang, Y., Liang, Y., et al. (2018). Permian paleogeography of the eastern CAOB: Paleomagnetic constraints from volcanic rocks in central eastern Inner Mongolia, NE China. *Journal of Geophysical Research: Solid Earth*, *123*(4), 2559–2582. https://doi.org/10.1002/2018jb015614

Zhang, D., Zhao, G., Huang, B., Zhao, J., & Zhao, Q. (2023). Palaeomagnetic constraints on a Late Permian to Early Triassic final closure of the Palaeo-Asian Ocean. *Geological Journal*, *58*(2), 903–919. https://doi.org/10.1002/gj.4633

Zhang, S., Gao, R., Li, H., Hou, H., Wu, H., Li, Q., et al. (2014). Crustal structures revealed from a deep seismic reflection profile across the Solonker suture zone of the Central Asian Orogenic Belt, northern China: An integrated interpretation. *Tectonophysics*, *612*, 26–39. https://doi.org/10.1016/j.tecto.2013.11.035

Zhou, T., Huang, B., Jia, S., Liang, Y., Zhang, D., Zhao, Q., et al. (2018). Paleomagnetic inclination shallowing in Lower Triassic Liujiagou Formation from Qinshui Basin, North China Block. *Beijing Da Xue Xue Bao*, *54*(3), 521–534.

References From the Supporting Information

Andersen, T. (2002). Correction of common lead in U–Pb analyses that do not report 204Pb. *Chemical Geology*, *192*(1–2), 59–79. https://doi.org/10.1016/s0009-2541(02)00195-x

Cogné, J. (2003). PaleoMac: A Macintosh™ application for treating paleomagnetic data and making plate reconstructions. *Geochemistry, Geophysics, Geosystems*, *4*(1). https://doi.org/10.1029/2001gc000227

19448007, 2023, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2023GL104881 by Curtin University Library, Wiley Online Library on [07/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Didenko, A., Li, Y.-F., Peskov, A. Y., Sun, S.-L., Karetnikov, A., & Zhou, Y.-H. (2016). Closure of the Solonker basin: Paleomagnetism of the Linxi and Xingfuzhilu formations (Inner Mongolia, China). *Russian Journal of Pacific Geology*, *10*(5), 317–336. https://doi.org/10.1134/s181971401605002x

Enkin, R. (1990). *Formation et déformation de l'Asie depuis la fin de lère primaire: Les apports de l'étude paléomagnétique des formations secondaires de Chine du Sud, Doctor Dissertation* (Vol. 7). University de Paris.

Fisher, R. A. (1953). Dispersion on a sphere. *Proceedings of the Royal Society of London - Series A: Mathematical and Physical Sciences*, *217*(1130), 295–305. https://doi.org/10.1098/rspa.1953.0064

Kirschvink, J. L. (1980). The least-squares line and plane and the analysis of palaeomagnetic data. *Geophysical Journal International*, *62*(3), 699–718. https://doi.org/10.1111/j.1365-246x.1980.tb02601.x

Liu, Y., Gao, S., Hu, Z., Gao, C., Zong, K., & Wang, D. (2010). Continental and oceanic crust recycling-induced melt–peridotite interactions in the Trans-North China orogen: U–Pb dating, Hf isotopes and trace elements in zircons from mantle xenoliths. *Journal of Petrology*, *51*(1–2), 537–571. https://doi.org/10.1093/petrology/egp082

Lowrie, W. (1990). Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties. *Geophysical Research Letters*, *17*(2), 159–162. https://doi.org/10.1029/gl017i002p00159

Ludwig, K. (2003). *User's manual for Isoplot/Ex version 3.00 e A geochronology toolkit for microsoft Excel* (Vol. 4). Berkeley Geochronological Center, Special Publication.

Tauxe, L., Shaar, R., Jonestrask, L., Swanson-Hysell, N., Minnett, R., Koppers, A., et al. (2016). PmagPy: Software package for paleomagnetic data analysis and a bridge to the magnetics information consortium (MagIC) database. *Geochemistry, Geophysics, Geosystems*, *17*(6), 2450–2463. https://doi.org/10.1002/2016gc006307

Zhao, X., Robert, S., Zhou, Y., Wu, H., & Wang, J. (1990). New paleomagnetic results from northern China: Collision and suturing with Siberia and Kazakhstan. *Tectonophysics*, *181*(1–4), 43–81. https://doi.org/10.1016/0040-1951(90)90008-v

19448007, 2023, 20, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2023GL104881 by Curtin University Library, Wiley Online Library on [07/01/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License